SOURAV KARAK

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PROFILE SUMMARY

- Results-driven professional with 4+ years of experience as a Teamcenter Developer delivering end-to-end PLM solutions using Siemens Teamcenter, complemented by 3+ years of experience in Full Stack Development
- Experienced in server-side and client-side customization using ITK, RAC, BMIDE, and Active Workspace (AWC)
- Hands-on experience in data model configuration, workflow automation, and working across multiple Teamcenter modules including Report Builder, Workflow Designer, Organization, Structure Manager, Classification, and Product Configurator
- Experienced in data migration from Teamcenter Engineering (TCE) to Teamcenter Unified Architecture (TCUA), ensuring data integrity and seamless transition
- Skilled in integrating Teamcenter with SAP S/4HANA for seamless PLM–ERP data exchange using RESTful APIs and cURL, including BOM synchronization and material master data integration
- Experienced in transferring Teamcenter business object metadata from TCUA to CATIA V6 (Dassault Systems), ensuring accurate cross-platform data consistency
- Proficient in developing and consuming SOAP-based web services for data transfer and integration with external enterprise systems
- Proficient in leveraging Teamcenter OOTB SOA services and developing custom SOA services for advanced Active Workspace customization
- Proven ability to troubleshoot and resolve complex issues, optimize system performance, and deliver high-quality solutions aligned with business requirements in manufacturing environments
- Effective team player with a strong passion for developing robust, efficient solutions aligned with business objectives, and a proven ability to thrive in collaborative team environments.

CORE COMPETENCIES

- **ITK, RAC, BMIDE, AWC** – Client & server-side customization
- **Business Objects & Data Model** – Business Objects, Attributes, LOV
- **Teamcenter Modules** – Report Builder, Workflow Designer, Organization, Structure Manager, Classification, Product Configurator
- **PLM-ERP Integration** – SAP S/4HANA, BOM synchronization, material master data exchange
- **Data Migration** – TCE → TCUA, metadata transfer TCUA to CATIA V6 (Dassault Systems)
- **SOA & Web Services** – OOTB/custom SOA, SOAP & REST APIs, cURL-based data transfer
- **Active Workspace Customization** – UI personalization, dashboarding, advanced AWC enhancements

SKILLS

Core Teamcenter Expertise

- Extensive experience in Siemens Teamcenter architecture, solution design, and enterprise-scale implementations
- Proficient with core Teamcenter tools: BMIDE, Workflow Designer, Structure Manager, Query Builder, Report Builder, Classifications, and Product Configurator
- Knowledge of Active Workspace (AWC) customization, extension, and UI configuration
- Familiar with the Mendix low-code platform

Customization & Development

- Develop and customize Teamcenter functionalities using ITK (Integration Toolkit), RAC (Rich Client), SOA (Service-Oriented Architecture), and advanced server-side customizations
- Hands-on experience with Handler, Exit, and custom workflow development
- Design, configure, and maintain the Teamcenter BMIDE (Business Modeling Independent Development Environment), including data modeling, object definitions, rules, and extensions

- Designed and implemented scalable SOAP and REST APIs for seamless communication between Teamcenter and external systems
- Experience in Active Workspace (AWC) customization and UI configuration
- Knowledge of integrating the Mendix low-code platform with Teamcenter Active Workspace (AWC)

Programming & Technologies

- Proficient in C, C++, and Java for Teamcenter customization and backend development
- Experience with JavaScript, HTML5, Bootstrap, jQuery, and CSS3 for Active Workspace (AWC) customization
- Skilled in HTML5, CSS3, XML, XSLT, and JSON for data transformation, UI, and configuration
- Experience with scripting languages (Python, Shell, and Batch) for automation and deployment
- Strong knowledge of database technologies, including Oracle and MySQL
- Experience in developing and consuming SOAP and REST web services for system integration (tools used: Postman, SoapUI and Boomerang)
- Experienced with version control systems such as Git

Integration & Enterprise Systems

- Designed and implemented automated data transfer from Teamcenter to SAP S/4HANA using cURL and JSON, ensuring accurate, real-time synchronization between PLM and ERP systems
- Integrated the Mendix low-code platform with Teamcenter Active Workspace (AWC) using REST/SOA services to enable seamless data exchange
- Integrated TCUA with CATIA V6 (Dassault Systems) to transfer metadata, ensuring accurate BOM, lifecycle, and version synchronization across PLM systems

Deployment, Architecture & DevOps

- Hands-on experience in Teamcenter installation and setup of development environments
- Executed BMIDE deployments for development environments, including data model updates and configuration management tasks
- Configured and optimized application servers (Tomcat) for development environments
- Configured Valgrind in SUSE Linux development environment to perform memory management analysis and debugging for Teamcenter ITK code
- Familiar with CI/CD pipelines and version control using Git for source code management and collaborative development

PROFESSIONAL EXPERIENCE

Tata Technologies Ltd || Senior Solution Developer

July, 2021 to Present

Project: TCE to TCUA Data Migration

- Identified the scope of migration, including data types (items, BOMs, CAD files, datasets, change objects), site-specific data, and users
- Analyzed existing TCE data models, identified gaps, and mapped them to TCUA Business Modeler IDE (BMIDE) models for accurate migration
- Developed and executed automated scripts (Shell) and tools (ITK Utility) to transfer metadata, BOMs, CAD files, and lifecycle information from TCE to TCUA
- Implemented post-migration verification processes using the TCE-to-TCUA data validation program (ITK Utility) to ensure consistency, completeness, and compliance with enterprise PLM standards
- Collaborated with cross-functional teams to troubleshoot migration issues and resolve data conflicts
- Documented migration methodology, best practices, and lessons learned for future PLM upgrades

Project: TCUA to SAP S/4HANA Data Transfer using cURL, CJSON, and REST Services

- Designed and implemented automated data transfer from Teamcenter Unified Architecture (TCUA) to SAP S/4HANA using cURL, CJSON, and REST services, ensuring accurate and real-time synchronization
- Developed scripts and integration workflows to transfer metadata, BOMs, and lifecycle information between TCUA and SAP S/4HANA via REST APIs
- Ensured data consistency and integrity during transfers by validating payloads and handling exceptions in integration scripts and REST calls
- Collaborated with SAP and Teamcenter teams to define API endpoints, mapping rules, and error-handling strategies for seamless PLM–ERP communication

- Optimized REST and cURL-based data transfer processes to reduce latency and improve performance across large datasets
- Documented integration methodology, scripts, REST endpoints, and best practices to support future PLM–ERP data exchange projects

Project: Dashboarding Teamcenter Modularity Data in Active Workspace (AWC)

- Designed and implemented interactive dashboards in Active Workspace (AWC) to visualize Teamcenter modularity data for engineering and management teams
- Developed custom reports in AWC and UI components using JavaScript, HTML5, Bootstrap, and CSS3 to display modularity metrics and lifecycle information
- Integrated Teamcenter REST APIs and SOA services to fetch real-time modularity data for dashboard visualization
- Configured filters, drill-downs, and reports to enable actionable insights on dashboards
- Collaborated with cross-functional teams to align dashboard metrics with business requirements and PLM processes
- Documented dashboard design, integration approach, and usage guidelines to support user adoption and future enhancement

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AB Softech Pvt Ltd || Software Developer
July, 2018 to June, 2021

Project: Full Stack Web Development

- Developed a dynamic full-stack web application using PHP and MySQL for backend and HTML, CSS, Bootstrap, jQuery for frontend
- Designed responsive and mobile-friendly user interfaces using Bootstrap, ensuring cross-browser compatibility
- Implemented asynchronous data handling using AJAX and jQuery, reducing page reloads and improving user experience
- Designed and managed relational database schemas in MySQL, ensuring data integrity and optimized query performance
- Validated and sanitized user inputs to prevent common vulnerabilities such as SQL injection and XSS
- Used Git/GitHub for version control and collaborative development

ACADEMIC PROJECTS & AWARDS

Project Title: Online application for college data management
Organization: Government College of Engineering & Textile Technology, Serampore
Contribution: Some layout designing and backend development
Description: College data management system gives a straightforward interface to view understudy data, staff information, attendance, fees, record etc.
Technologies: ASP.NET, MySQL, HTML/CSS, Bootstrap, JavaScript, Ajax, jQuery
Developer Tools: Notepad++, Visual Studio, VS Code, Sublime Text, GIT
Year: 2018

Award Title: Demonstration of "Agility" in dedicated work on improving CV Modularity data in AWC
Organization: Tata Motors
Description: Indian Copyright Application in the form of notarization –
Sr. No. MH/PUNE/691/277/23. For vehicle DMU creation using t-matrix in Tata Motors.
Year: 2023

Award Title: Best Project Team of the year
Organization: AB Softech PVT LTD.
Description: For delivering most no of projects in a financial year (2019 – 2020)
Year: 2020

EDUCATION

- Bachelor of Technology in Computer Science and Engineering from Government College of Engineering & Textile Technology, Serampore-WB, India.